# Technical Document #843: Cloud Computing - Comprehensive Overview

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Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

### Key Takeaways:

- Content delivery networks (CDNs) distribute content globally to reduce latency.
- Microservices architecture structures applications as a collection of loosely coupled services.
- Kubernetes orchestrates containerized applications across clusters of machines.